



Maintenance Schedule

Diesel engine

16V/20V1163Mx4

Application group 1B 1D 1DS

MS50111/04E

Engine model	kW/cyl.	Application group
16V1163M74	300 kW/cyl.	1B; continuous operation, variable, high load
20V1163M74	300 kW/cyl.	1B; continuous operation, variable, high load
16V1163M84	325 kW/cyl.	1D; continuous operation, variable, medium load
20V1163M84	325 kW/cyl.	1D; continuous operation, variable, medium load
16V1163M94	370 kW/cyl.	1DS; continuous operation, variable, low load
20V1163M94	370 kW/cyl.	1DS; continuous operation, variable, low load

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 TBO matrix - Application groups 1B, 1D, 1DS

16V/20V1163Mx4							
Load profile		Load factor %	Load indicator %	0 to 300 kW/cyl.	301 to 325 kW/cyl.	326 to 370 kW/cyl.	
Load %	Time %						
110 ₍₁₁₀₋₁₀₀₎	—	0 to 38	≤ 3	30,000	27,000	24,000	—
100 ₍₁₀₀₋₉₀₎	1						
90 ₍₉₀₋₈₀₎	—						
80 ₍₈₀₋₆₅₎	4						
65 ₍₆₅₋₅₀₎	—						
50 ₍₅₀₋₃₀₎	40						
30 ₍₃₀₋₅₎	45						
Idle speed ₍₅₋₀₎	10						
110 ₍₁₁₀₋₁₀₀₎	—	39 to 62	≤ 12	27,000	24,000	18,000	—
100 ₍₁₀₀₋₉₀₎	10						
90 ₍₉₀₋₈₀₎	—						
80 ₍₈₀₋₆₅₎	5						
65 ₍₆₅₋₅₀₎	65						
50 ₍₅₀₋₃₀₎	—						
30 ₍₃₀₋₅₎	20						
Idle speed ₍₅₋₀₎	—						
110 ₍₁₁₀₋₁₀₀₎	—	63 to 73	≤ 22	24,000	21,000	12,000	—
100 ₍₁₀₀₋₉₀₎	3						
90 ₍₉₀₋₈₀₎	40						
80 ₍₈₀₋₆₅₎	42						
65 ₍₆₅₋₅₀₎	—						
50 ₍₅₀₋₃₀₎	—						
30 ₍₃₀₋₅₎	—						
Idle speed ₍₅₋₀₎	15						
110 ₍₁₁₀₋₁₀₀₎	—	74 to 81	≤ 31	21,000	18,000	10,000	—
100 ₍₁₀₀₋₉₀₎	10						
90 ₍₉₀₋₈₀₎	50						
80 ₍₈₀₋₆₅₎	30						
65 ₍₆₅₋₅₀₎	—						
50 ₍₅₀₋₃₀₎	—						
30 ₍₃₀₋₅₎	5						
Idle speed ₍₅₋₀₎	5						

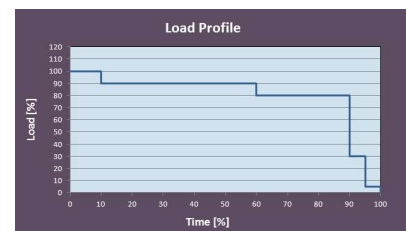
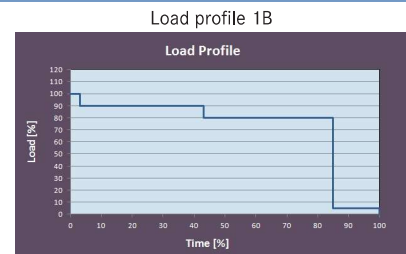
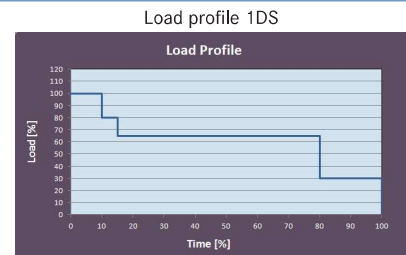


Table 2: TBO matrix - Application groups 1B, 1D, 1DS

Application 1B:

Max. number of load cycles 1 (<10% to >90%):2/H

Max. number of load cycles 2 (<10% to >70...90%):2/H

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Application 1D/DS:

Max. number of load cycles 1	(<10% to >90%):0.5/H
Max. number of load cycles 2	(<10% to >70...90%):2/H

1.4 Maintenance Schedule Matrix 30000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00321	1 DAY																														
WM00126	1 YR																														
WM00128	1 YR																														
WM00313	2 YR																														
WM01285	1 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00086	1 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00194	1 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00069	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00094	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X					X						X						X							X
WM00136	1 YR						X					X						X					X								X
WM00076	2 YR										X										X										X
WM00090	18 YR										X										X										X
WM00110	18 YR										X										X										X
WM01287	18 YR										X										X										X
WM01288	18 YR										X										X										X
WM01289	18 YR										X										X										X
WM01291	18 YR										X										X										X
WM00376	18 YR																				X										
WM00377	18 YR																				X										
WM00970	18 YR																				X										
WM01292	18 YR																				X										
WM01293	18 YR																				X										
WM00003	18 YR																														X
WM00062	18 YR																														X
WM00099	18 YR																														X
WM00100	18 YR																														X
WM00101	18 YR																														X
WM00108	18 YR																														X
WM00164	18 YR																														X
WM00315	18 YR																														X
WM00367	18 YR																														X
WM00369	18 YR																														X
WM00378	18 YR																														X
WM00381	18 YR																														X
DAY = Days																															
YR = Years																															
MON = Months																															
HRS = Hours																															

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Task	Limit	Operating hours [HRS]																	
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000
WM00974	18 YR																		
WM00980	18 YR																		
WM00996	18 YR																		
WM01290	18 YR																		
WM01294	18 YR																		
WM01295	18 YR																		
WM01296	18 YR																		
WM01297	18 YR																		
WM01298	18 YR																		
WM01299	18 YR																		
WM00439	18 YR																		
WM00978	18 YR																		
WM00001	18 YR																		
WM00002	18 YR																		
WM00011	18 YR																		
WM00074	18 YR																		
WM00077	18 YR																		
WM00088	18 YR																		
WM00091	18 YR																		
WM00095	18 YR																		
WM00103	18 YR																		
WM00111	18 YR																		
WM00127	18 YR																		
WM00137	18 YR																		
WM00172	18 YR																		
WM01300	18 YR																		
WM01301	18 YR																		
WM01306	1 MON																		
WM01307	3 YR																		
WM01308	18 YR																		
WM01222	1 YR		X		X		X		X		X		X		X		X		X
WM01055	1 YR																		
WM00148	3 YR																		
WM00979	18 YR																		
WM01302	1 YR		X		X		X		X		X		X		X		X		X
WM00124	3 YR					X				X				X			X		X
WM01576	18 YR																		
WM00881	1 MON																		
WM00925	18 YR																		
WM01578	18 YR																		
WM00638	1 MON		X		X		X		X		X		X		X		X		X
WM00639	1 MON		X		X		X		X		X		X		X		X		X
WM00642	1 MON		X		X		X		X		X		X		X		X		X
WM01208	1 MON		X		X		X		X		X		X		X		X		X
WM01303	1 YR		X		X		X		X		X		X		X		X		X
DAY = Days																			
YR = Years																			
MON = Months																			
HRS = Hours																			

Task	Limit	Operating hours [HRS]																			
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000
WM01304	1 YR																				X
WM01305	18 YR																				
WM00790	1 DAY																				
WM00017	1 MON		X		X		X		X		X		X		X		X		X		X
WM00015	1 YR		X		X		X		X		X		X		X		X		X		X
WM00140	1 YR		X		X		X		X		X		X		X		X		X		X
WM00013	2 YR		X		X		X		X		X		X		X		X		X		X
WM00215	1 DAY																				
WM00216	1 DAY																				
WM00214	1 YR		X		X		X		X		X		X		X		X		X		X
WM01577	2 YR									X								X			X
DAY = Days YR = Years MON = Months HRS = Hours																					

15500 to 30000 Operating hours

Task	Limit	Operating hours [HRS]																			
		15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000	24500	25000
WM00285	1 DAY																				
WM00286	1 DAY																				
WM00287	1 DAY																				
WM00288	1 DAY																				
WM00289	1 DAY																				
WM00321	1 DAY																				
WM00126	1 YR																				
WM00128	1 YR																				
WM00313	2 YR																				
WM01285	1 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00086	1 YR		X		X		X		X		X		X		X		X		X		X
WM00194	1 YR		X		X		X		X		X		X		X		X		X		X
WM00069	2 YR		X		X		X		X		X		X		X		X		X		X
WM00094	2 YR		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X				X						X				X
WM00136	1 YR						X				X						X				X
WM00076	2 YR									X							X				X
WM00090	18 YR									X							X				X
WM00110	18 YR									X							X				X
WM01287	18 YR									X							X				X
WM01288	18 YR									X							X				X
WM01289	18 YR									X							X				X
WM01291	18 YR									X							X				X
WM00376	18 YR									X											X
DAY = Days YR = Years MON = Months HRS = Hours																					

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